
STAGE 1
Raw Dataset Ingestion
BankChurners ($N = 10,127$)
13 Continuous, 6 Categorical
Target: Attrition_Flag (16%)
STAGE 2
Feature Hygiene
Pruning Artifacts
Drop CLIENTNUM & NB artifacts
Final Predictors: $D = 19$
STAGE 3
Stratified Splitting
5 Outer Folds ($K_{\text{out}} = 5$)
Train: 80% ($N_{\text{tr}} = 8,101$)
Val: 20% ($N_{\text{val}} = 2,026$)
STAGE 4
Train Preprocessing
Fit Exclusively on Train
One-Hot & z-Score (μ_j, σ_j)
Strict Zero-Leakage Fit
STAGE 5
MCAR Injection
Sparsity $\rho \in [15\%, 90\%]$
Binary mask $\mathbf{M} \in \{0,1\}^{N \times D}$
Feasibility: ≥ 1 obs/row
STAGE 6
Dual Holdout Eval
Quarantined Testing
Clean: pristine $\mathbf{X}_{\text{val}}$
Incomp: corrupted $\widetilde{\mathbf{X}}_{\text{val}}$